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Procedia Computer Science 260 (2025) 415–423

Procedia Computer Science

www.elsevier.com/locate/procedia

Seventh International Conference on Recent Trends in Image Processing and Pattern Recognition (RTIP2R-2024)

Deep Facial Feature Fusion and Voting Strategies for Enhanced Emotion Recognition

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Abstract

This paper presents a novel approach that enhances emotion recognition by leveraging deep facial feature fusion and optimized voting strategies. Unlike conventional methods that rely on a single type of feature or classifier, our approach integrates feature fusion in deep learning architecture. We employ a fusion mechanism that combines features at multiple levels, enabling a more comprehensive representation of emotional cues. Additionally, a voting strategy is introduced to refine the final emotion classification, effectively reducing the impact of misclassifications and improving overall accuracy. The proposed system is rigorously evaluated on benchmark dataset, demonstrating its superior performance compared to state-of-the-art methods. The experimental results show that our approach not only achieves higher accuracy but also exhibits robustness across varying facial expressions, lighting conditions, and occlusions.

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Peer-review under responsibility of the scientific committee of the Seventh International Conference on Recent Trends in Image Processing and Pattern Recognition.

Keywords: Facial Expression; Feature Fusion; Emotion Recognition; Voting; Deep Net

1. Introduction

Significant advancements in FER have been witnessed recently, owing to state of the art techniques from the fields of deep learning. [1]. Feature fusion plays a crucial role in FER by integrating diverse information extracted from multiple sources or modalities. In this context, feature fusion combines various types of features, such as geometric features [2], appearance-based features [3], and spacial temporal features [4], to enhance the discriminative power of the recognition system. [5] utilized Multi-channel deep spatial temporal feature fusion neural network (MDSTFN)

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10.1016/j.procs.2025.03.218

to extract and fuse the temporal and spatial features of facial expressions in a static image. A multichannel network is proposed by Sun et al. [5], which extracted spatial information from facial expression and temporal information (optical flow) from the changes between emotional and neutral faces. This paper introduces a novel framework for enhanced emotion recognition by exploiting deep facial feature fusion from VGG19, dimensionality reduction using Principal Component Analysis (PCA), and innovative voting strategies among multiple classifiers. Specific contributions are the following :

1. The VGG19 deep learning model is used. VGG19 is renowned for its effectiveness in facial image processing, to extract high-dimensional facial features. This study exploits the high level feature extraction capability of VGG19 for further processing.
2. To address the challenges of high-dimensional data, PCA is applied for feature dimensionality reduction. This step enhances the computational efficiency and reduces the risk of overfitting by retaining the most significant features that contribute to emotion recognition. This approach customizes PCA application to preserve the emotional salience of facial features, thereby ensuring that the essential characteristics for emotion discrimination are maintained.
3. A significant contribution of this framework is the implementation of a strategic voting mechanism among multiple classifiers. This method improves the robustness and accuracy of emotion recognition by aggregating predictions from diverse classifiers. Various voting techniques are explored and optimized, including majority voting, weighted voting, and dynamic rule-based voting, to determine the most effective strategy for this specific application.

Our research introduces a novel framework that significantly enhances emotion recognition by leveraging deep facial feature extraction using VGG19, dimensionality reduction through Principal Component Analysis (PCA), and advanced voting strategies among multiple classifiers. Unlike the Differential Geometric Fusion Network (DGFN) in [6], which combines handcrafted and deep features, our approach focuses on the sophisticated VGG19 model for high-dimensional feature extraction, coupled with PCA to improve computational efficiency and reduce overfitting. In contrast to the lightweight Multi-feature Fusion Based CNN (MFF-CNN) in [7], which extracts global and local features, our method prioritizes essential feature preservation and incorporates a robust voting mechanism for improved accuracy and robustness. Finally, compared to the hybrid deep learning model in [8], which integrates spatial and temporal features, our approach stands out by combining deep feature extraction, dimensionality reduction, and sophisticated voting techniques to address class imbalances and achieve a high F1-score of 98% in facial expression recognition. The paper is structured as follows: Section II provides a brief overview of related work on feature extraction techniques involving both handcrafted features and deepnet features. Section III outlines the proposed method, detailing base model training, feature extraction technique, and classification of facial expression using extracted features projected on eigen feature space. Section IV discusses the CK+ dataset, including preprocessing, encountered problems, and its usage. Section V presents experimental results, and Section VI concludes the paper.

2. Review Work

Facial Emotion Recognition (FER) involves three primary stages [9]: initial preprocessing, which entails face detection; followed by feature extraction or selection, the focus of this study; and finally, classification. In [10], 3D Convolution is utilized for spatiotemporal feature extraction, which is subsequently cascaded with Deep Belief Network (DBN) for prediction. In [11], C3D is employed as a feature extractor, with a network of vector of Locally Aggregated Descriptors (NetVLAD) layer utilized to aggregate temporal information of motion features by learning cluster centers. In [12] Deep features are proposed to be extracted from various facial parts derived from texture and depth images, subsequently fused together to interconnect them with feedback.

Several studies are conducted on FER using deep convolutional neural network (DCNN). Chirra et al. [13] proposed a multi-block DCNN model aimed at extracting discriminative features for recognizing seven facial emotions of virtual characters, utilizing three classifiers: SVM, Random Forest (RF), and Logistic Regression (LR). Shuo Cheng et al. [14] emphasized the effectiveness of DCNN in feature extraction compared to shallow CNN, stressing the importance of determining optimal layer numbers to prevent overfitting. Qi Zhang [15] explored FER using Local Binary Pattern

(LBP) in conjunction with the Visual Geometry Group (VGG) network, achieving superior accuracy compared to traditional methods. The application of pre-trained CNN, particularly the VGG19 network, for FER, demonstrating improved classification accuracy through feature transfer is investigated in [16] . Transfer learning is utilized in [17]on AlexNet and VGG19 models combined with SVM classifier, achieving high accuracy rates in FER. Denys V. P. et al. [18] explored CNN for automated FER and emotion detection, advocating for transfer learning to enhance system performance, particularly in scenarios with limited labeled data.

3. Proposed methodology

3.1. Motivation

Deep neural networks (DNNs), especially Convolutional Neural Networks (CNNs), outperform traditional handcrafted features in computer vision tasks by automatically learning hierarchical representations from raw data. VGG19 [19], a well-known DNN architecture, is particularly effective for facial expression recognition (FER) due to its ability to learn deep, transferable features and its robustness to facial variations. Pre-trained on large datasets like ImageNet, VGG19 efficiently extracts features across different levels, from simple edges to complex semantic information, making it ideal for FER applications. This attribute is exploited in this study to capture the higher level features of different facial expressions. Motivating with the above advantages, in this study VGG19 is used for deep feature extraction.

3.2. Building Base Model And Training

The model construction begins with selecting VGG19 architecture, pretrained on ImageNet dataset due to its robustness in image classification. ImageNet, a diverse dataset with 14+ million images across 20,000+ categories. The top fully connected layers (f_c) are removed and prepare the model for transfer learning, while freezing the weights learned by the convolution layers. Global Average Pooling (GAP) layer is done for reducing parameter space, followed by a f_c for classification tasks. Two different models are built as shown in in Fig.(1) ; the first one is for 8 classes scenario and another one for classifying 7 classes, having 4104 and 3591 number of trainable parameters respectively.

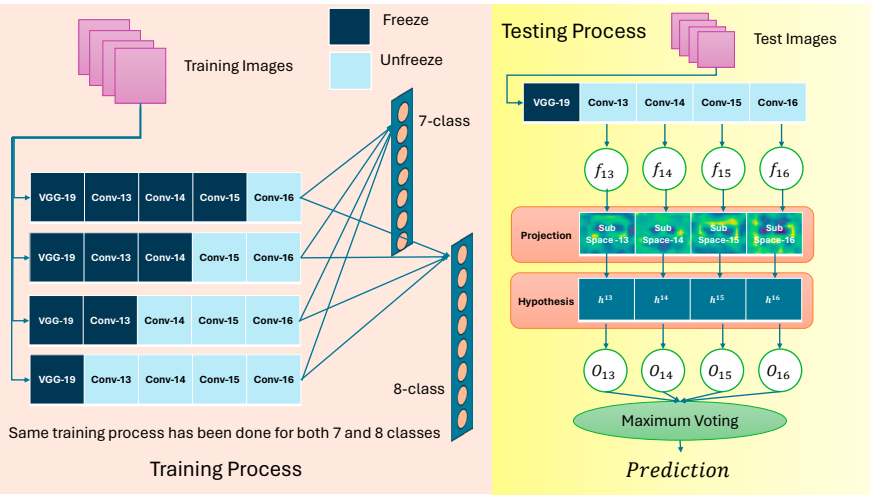


Fig. 1: The training images are input into a pre-trained deep network with specific layers frozen for fine-tuning, producing a feature extractor. Features from layers $l = 13, 14, 15, 16$ are independently extracted. For each test image, four features are extracted and projected onto respective eigen-feature spaces. The projected feature dimensions are 1073, 1193, 952, and 1537 for the dataset including the neutral class, and 1141, 1289, 1042, and 1623 for the dataset excluding the neutral class. These features are fed into hypotheses $h_l^{(k)}$, where k denotes the hypothesis type (e.g., SVM, KNN). Class labels are generated, and maximum voting determines the final prediction

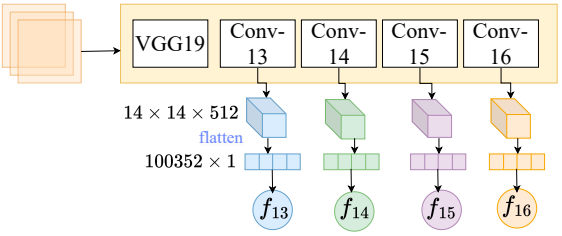


Fig. 2: The figure shows the feature extraction from the fine-tuned VGG19 architecture. From layer l, feature maps are extracted having an output volume of dimension $d_h \times d_w \times c$, where each of the c channels have height d_h and width d_w. These extracted volumes are flattened for further processing.

3.3. Layerwise Independent Fine Tuning And Feature Extraction

In the subsequent model refinement phase, layer wise independent fine-tuning are employed for enhanced performance and high-level feature extraction. At first, from layer l, where ($l = 16, 15, 14, 13$) to the f_c layer are unfreezed and used for fine-tuning while rest of the layers are freezed. This process is iterated until the process achieved layer $l = 13$. To extract features the flatten layer is used to convert the convolution volumes as shown in Fig. 2 . The feature maps obtained with this mechanism termed as F_l, where ($l = 16, 15, 14, 13$).

F_l, the feature map for all m images having dimension of $n_l \times m$, where

$$n_l = d_h \times d_w \times c = 100352$$

$d_h = 14$ (height of feature map), $d_w = 14$ (width of feature map), $c = 512$ (no. of channels) and m represents the number of input images and each of them is an n_l dimensional vector which represents the flattened representation of the extracted features in the high-level feature vector space.

This iterative fine-tuning and feature extraction process refines the model to better fit the CK+ dataset while efficiently extracting relevant high-level features, optimizing computational resources.

3.4. Feature Representation

$$F_l = \begin{bmatrix} \vdots & \vdots & \vdots & & \vdots \\ f_l^{(1)} & f_l^{(2)} & f_l^{(3)} & \cdots & f_l^{(m)} \\ \vdots & \vdots & \vdots & & \vdots \end{bmatrix}_{n_l \times m} \tag{1}$$

The matrix F_l represents the high-level features extracted from convolution layer l for each input image, each having dimension $n_l \times m$. These features are extracted by passing the sample image to the base model, and then flattening the high-level features generated by layer l, as shown in Fig. 2 .

$$f_l^{(i)} = \begin{bmatrix} f_{l1}^{(i)} & f_{l2}^{(i)} & \cdots & f_{ln_f}^{(i)} \end{bmatrix}^T_{1 \times n_f} \tag{2}$$

$f_{lj}^{(i)}$ is the jth feature, extracted from layer l for sample image i.

3.5. ML Independent Model Training

F is aggregated feature matrix of dimension $m \times n$ where $n = 4 \times n_l = 401408$ each row corresponding to a sample image and each set of consecutive n features corresponding to features extracted from convolution layer l.

$$F = \left[F_{16}^T \; F_{15}^T \; F_{14}^T \; F_{13}^T\right]_{m \times n} \tag{3}$$

After extracting features, the F is standardized , ensuring zero mean and unit variance. Following this, Principal Component Analysis (PCA) is applied on each feature space F_l to reduce dimensionality while retaining 95% of the variance, projecting the feature space to a lower-dimensional eigen feature vector space. Subsequently, independent machine learning models such as SVM, RF, k-Nearest Neighbors (KNN) and LR are trained using the eigen features for each layer. By training separate models on these features, a diverse set of classifiers is constructed, each exploiting different aspects of the data for classification tasks.

This approach enables the exploration of various classifiers and their performance on the eigen feature space, providing insights into which models are most effective for the specific task at hand. It also allows for comparison and selection of the best-performing model or combination of models, ultimately leading to improved classification accuracy and robustness.

$$\hat{O}_l = \left[\ldots \; \hat{O}_l^{(i)} \; \ldots\right]_{1 \times m}^T \tag{4}$$

$\hat{O}_l^{(i)}$ is the output of the machine learning model trained spcifically on the eigen features extracted from the layer l, whereas i denoting the corresponding sample image

3.6. Prediction System using Maximum Voting

$$predction^{(i)} = \max_l count(\hat{O}_l^{(i)}) \tag{5}$$

In the prediction system utilizing Maximum Voting given in Fig.3, the focus lies on combining predictions from similar machine learning models trained on features extracted from each layer.

The process involves each individual model generating prediction $\hat{O}_l^{(i)}$ for a given image. The final prediction for the image is determined by aggregating these individual predictions and selecting the class with the highest frequency, essentially employing a voting mechanism. The class that receives the most votes across all models is chosen as the final prediction.

By merging predictions from multiple models, the system leverages diverse insights captured by each model. This ensemble approach enhances prediction robustness and accuracy by mitigating biases or limitations inherent in any single model. It effectively combines the strengths of various models to provide more reliable predictions, especially beneficial in complex datasets or tasks with distinguished patterns.

4. Dataset

The Extended Cohn-Kanade (CK+)[20] database is a standard dataset for evaluating Facial Expression Recognition (FER) systems, consisting of 593 video sequences from 123 subjects, with 327 sequences labeled with seven basic expressions based on FACS . After excluding neutral classes, the dataset comprises 5885 images, split into 4708 for

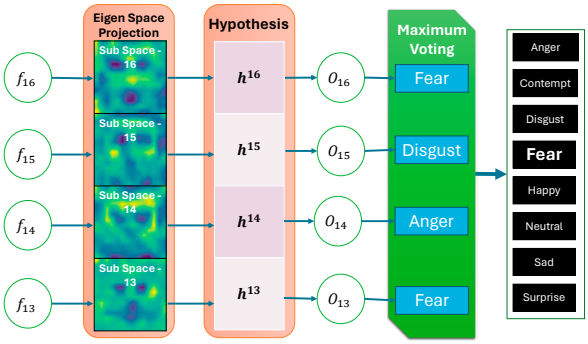


Fig. 3: Maximum Voting Classifier

training and 1177 for testing (80/20 split). Including neutral classes, there are 6213 images, with 4971 for training and 1242 for testing. The dataset is imbalanced, particularly in the contempt, sad, and neutral classes, which are underrepresented.

5. Experimental Results And Analysis

5.1. Parameter Configuration and Setup

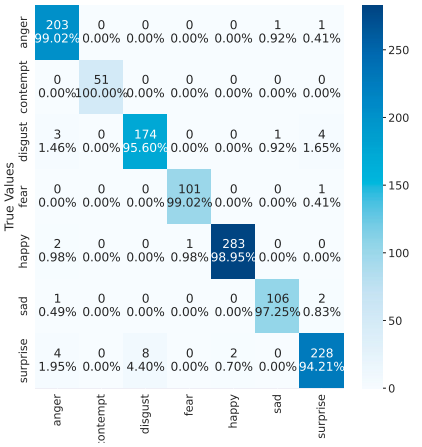
The base model was trained with a learning rate of 0.0001 using sparse categorical crossentropy loss over 30 epochs with a batch size of 64. Fine-tuning followed with a reduced learning rate of 0.00001 for 20 epochs, maintaining the same batch size. Adam optimizer was used with $\beta_1 = 0.9$, $\beta_2 = 0.999$, and $\epsilon = 1e^{-7}$. Extracted features were projected onto eigenspaces and classified using predefined configurations: SVM with an RBF kernel and $C = 1.0$, KNN with 5 neighbors using Euclidean distance, LR with a regularization strength of 3 and $C = 1.0$, and RF with 100 trees using the Gini criterion. Facial regions were cropped, converted to grayscale, and resized to 244×244 pixels to focus on essential facial features while removing irrelevant background information.

5.2. Key Findings

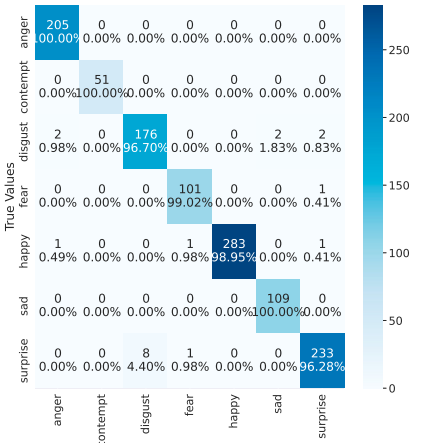
Test images were captured and passed through a pretrained DCNN model. Features were extracted from layer l independently. Each test image resulted in four sets of features, which were then projected onto four eigen-feature spaces. The projected weights obtained for the dataset including neutral class were 1073, 1193, 952, and 1537, while for the dataset excluding neutral class, they were 1141, 1289, 1042, and 1623. These weights were fed into a trained model, denoted as $h^{(k)}$, where k represents hypothesis types such as SVM, KNN, LR and RF . The voting scheme was employed to determine the final class label by considering the maximum votes from all $h^{(k)}$. Receiver Operating Characteristic (ROC) plots for SVM, KNN, LR and RF were plotted based on the maximum voting. Confusion matrices were generated to assess the performance of the voting scheme using fused extracted features from the fine-tuned model. A table provided the efficacy of the proposed method, evaluated using Precision, Recall, and F1-Score, with F1-score chosen due to class imbalance in the dataset. Table 1, showcased a 98% F1-score in FER.

Fig.4 displays the confusion matrices of classifiers trained on the CK+ dataset excluding neutral class. The matrices illustrate the performance of each model in predicting facial expressions on the test set, specifically focusing on emotions such as happiness, sadness, anger, surprise, fear, and disgust.

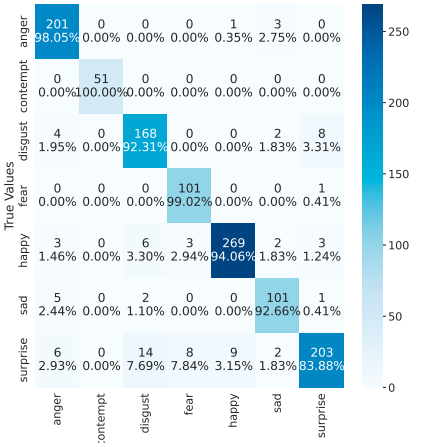
Table 1 presents the classification report of various classifiers tested on the CK+ dataset, with the exclusion of the neutral class. The classifiers, namely SVM, KNN, LR, and RF, are evaluated based on precision, recall, and F1-score metrics. Table 1 indicates that the classifiers used for voting scheme high precision, recall, and F1-scores, with values close to 1, demonstrating their effectiveness in FER on the dataset. Table 2 presents the classification report for various classifiers tested on the CK+ dataset, including the neutral class. In this table, LR show good precision, recall, and F1-scores across the board, suggesting a good performance. In contrast, with the inclusion of the neutral class, the



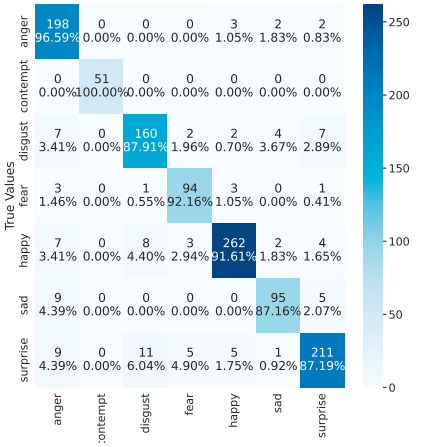
(a) Random Forest



(b) Logistic Regression



(c) KNN



(d) SVM

Fig. 4: Confusion Matrices of clssifiers trained on CK+ dataset (excluding neutral class).

Table 1: Classification report of hypothesis tested on CK+ dataset (excluding neutral class)

Classifier (Using Maximum Voting)	Precision	Recall	F1-Score	Accuracy
SVM	0.91	0.91	0.91	0.91
KNN	0.93	0.93	0.93	0.93
Logistic Regression	**0.98**	**0.98**	**0.98**	**0.98**
Random Forest	0.97	0.97	0.97	0.97

classifiers perform poor specifically on the neutral class only, which can inferred from the difference in the values of Table 1 and Table 2. Overall, the inclusion of the neutral class seems to have a significant impact on the performance of all classifiers.

Table 2: Classification report of hypothesis tested on CK+ dataset (including neutral class)

Classifier (Using Maximum Voting)	Precision	Recall	F1-Score	Accuracy
SVM	0.81	0.86	0.83	0.86
KNN	0.83	0.87	0.85	0.87
Logistic Regression	**0.89**	**0.88**	**0.88**	**0.88**
Random Forest	0.87	0.88	0.88	0.88

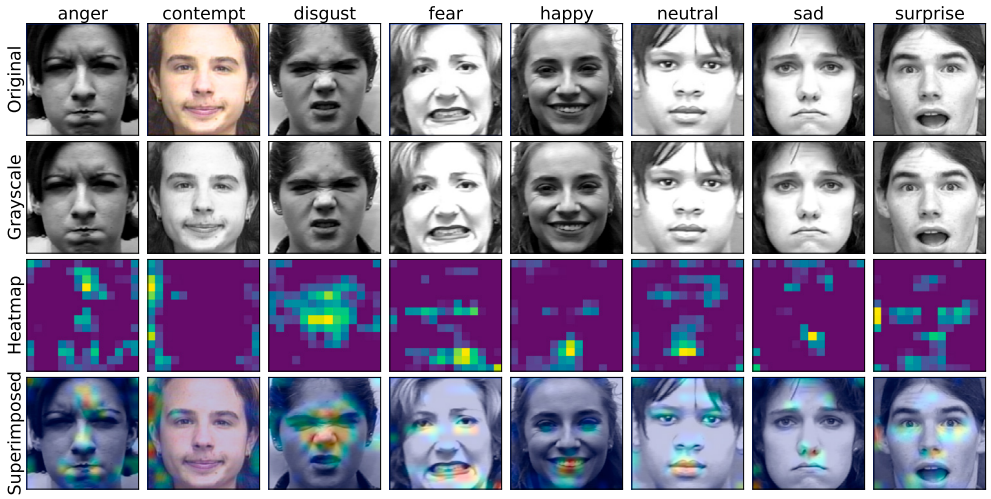


Fig. 5: GradCam images from finetuned model for each class.



(a) Original: happy Predicted: disgust

(b) Original: happy Predicted: fear

(c) Original: sad Predicted: surprise

(d) Original: surprise Predicted: disgust

Fig. 6: Images of false predictions.

In Fig.5 , GradCAM highlighted regions surrounding the eyebrows and eyes for surprise, with high activation indicative of their importance in recognition. Similarly, regions around the eyes and mouth are emphasized for happiness, and around the mouth and eyebrows for sadness, with high activation demonstrating their significance. For anger, GradCAM focused on regions around the eyebrows, eyes, and mouth, reflecting their relevance in recognition. Fear expressions prompted GradCAM to highlight regions around the eyes and eyebrows, while for disgust, emphasis is placed on regions around the nose and mouth. Contempt is characterized by highlighting regions around the mouth and eyes. In contrast, for neutral expressions, GradCAM showed relatively low activation across the entire face, reflecting the absence of strong emotional cues. Figure 6 shows the false predictions from the maximum voting classifiers. Mostly these images are very close to the neutral class, which making the classifier confused whether person is going to show neutral expression or any other expression, as there are images in the dataset which are actually non-neutral but are very close to the neutral class.

6. Conclusion, Limitations and Future Work

This study demonstrates that excluding neutral classes improves FER results, with ML models like SVM, KNN, RF, and LR achieving an F1-score of 98%. Including the neutral class presents challenges, reducing the F1-score to 88%. Fine-tuning significantly decreases training time and the number of trainable parameters in the CNN model. Maximum voting helps mitigate layer-specific biases, enhancing prediction reliability. The proposed framework lacks end-to-end training, requiring separate stages for feature extraction, dimensionality reduction, and classification. SVMs with non-linear kernels are not GPU-trainable, limiting scalability and slowing training, especially with large datasets or high-dimensional features. Additionally, including a neutral class complicates classification due to its overlap with other emotional classes. These factors highlight areas for potential improvement and future research directions.

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